

Absolute Metrics Tables

task_0

Table 1: Absolute metrics (task_0)

Model	Task 0	Task 1	Task 2	Task 3	Task 4	TRN AVG	HLD AVG
Base (old)	0.60	0.28	0.78	0.82	0.29	0.55	0.20
Base	0.66 ± 0.02	0.20 ± 0.02	0.89 ± 0.01	0.65 ± 0.01	0.32 ± 0.02	0.54 ± 0.01	0.40 ± 0.00
GRPO task_0 train={bs:8192,steps:10} eval=80	0.65	0.17	0.75	0.64	0.24	0.49	0.40
GRPO task_0 train={bs:8192,steps:10} eval=80	0.79	0.23	0.82	0.57	0.20	0.52	0.38
GRPO task_0 train={bs:8192,steps:10} eval=80	0.65	0.17	0.75	0.64	0.24	0.49	0.40
GRPO task_0 train={bs:8192,steps:10} eval=80	0.79	0.23	0.82	0.57	0.20	0.52	0.38
GRPO task_0 train={bs:2048,steps:50} eval=800	0.85	nan	nan	nan	nan	0.85	nan
GRPO task_0 train={bs:4096,steps:50} eval=80	0.90	0.17	0.94	0.68	0.26	0.59	0.39

task_1

Table 2: Absolute metrics (task_1)

Model	Task 0	Task 1	Task 2	Task 3	Task 4	TRN AVG	HLD AVG
Base (old)	0.60	0.28	0.78	0.82	0.29	0.55	0.20
Base	0.66 $\pm$ 0.02	0.20 $\pm$ 0.02	0.89 $\pm$ 0.01	0.65 $\pm$ 0.01	0.32 $\pm$ 0.02	0.54 $\pm$ 0.01	0.40 $\pm$ 0.00
SFT teacher eval (task 1)	0.00	0.95	0.00	0.00	0.00	0.19	0.00
GRPO task_1 train={bs:8192,steps:20} eval=320	0.62	0.27	0.94	0.79	0.30	0.59	0.35
GRPO task_1 train={bs:8192,steps:20} eval=320	0.68	0.38	0.91	0.64	0.34	0.59	0.42
GRPO task_1 train={bs:2048,steps:50} eval=800	nan	0.37	nan	nan	nan	0.37	nan
GRPO task_1 train={bs:8192,steps:20} eval=320	0.62	0.27	0.94	0.79	0.30	0.59	0.35
GRPO task_1 train={bs:8192,steps:20} eval=320	0.68	0.38	0.91	0.64	0.34	0.59	0.42
GRPO task_1 train={bs:8192,steps:40} eval=320	0.65	0.61	0.90	0.64	0.38	0.64	0.44
GRPO task_1 train={bs:8192,steps:60} eval=320	0.67	0.75	0.89	0.68	0.44	0.69	0.43
GRPO task_1 train={bs:4096,steps:50} eval=80	0.91	0.76	0.81	0.70	0.35	0.71	0.41
OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.69	0.31	0.70	0.72	0.42	0.57	0.34
OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.62	0.21	0.78	0.78	0.34	0.55	0.19
OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.57	0.46	0.70	0.68	0.09	0.50	0.13
OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.69	0.31	0.70	0.72	0.42	0.57	0.34
OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.62	0.21	0.78	0.78	0.34	0.55	0.19
OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.57	0.46	0.70	0.68	0.09	0.50	0.13
OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.69	0.31	0.70	0.72	0.42	0.57	0.34
OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.62	0.21	0.78	0.78	0.34	0.55	0.19
OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.57	0.46	0.70	0.68	0.09	0.50	0.13
OPD teacher=RL task_1 train={bs:8192,steps:50} eval=80	0.70	0.61	0.70	0.81	0.31	0.63	0.34

task_4

Table 3: Absolute metrics (task_4)

Model	Task 0	Task 1	Task 2	Task 3	Task 4	TRN AVG	HLD AVG
Base (old)	0.60	0.28	0.78	0.82	0.29	0.55	0.20
Base	0.66 ± 0.02	0.20 ± 0.02	0.89 ± 0.01	0.65 ± 0.01	0.32 ± 0.02	0.54 ± 0.01	0.40 ± 0.00
SFT teacher eval (task 4)	0.01	0.00	0.09	0.15	0.87	0.22	0.02
GRPO task_4 train={bs:8192,steps:20} eval=320	0.34	0.08	0.88	0.54	0.33	0.43	0.31
GRPO task_4 train={bs:8192,steps:20} eval=320	0.53	0.28	0.87	0.72	0.61	0.60	0.39
GRPO task_4 train={bs:8192,steps:40} eval=320	0.40	0.11	0.84	0.60	0.57	0.50	0.34
GRPO task_4 train={bs:8192,steps:40} eval=320	0.53	0.29	0.88	0.72	0.70	0.62	0.41
GRPO task_4 train={bs:8192,steps:20} eval=320	0.34	0.08	0.88	0.54	0.33	0.43	0.31
GRPO task_4 train={bs:8192,steps:20} eval=320	0.53	0.28	0.87	0.72	0.61	0.60	0.39
GRPO task_4 train={bs:8192,steps:40} eval=320	0.40	0.11	0.84	0.60	0.57	0.50	0.34
GRPO task_4 train={bs:8192,steps:40} eval=320	0.53	0.29	0.88	0.72	0.70	0.62	0.41
GRPO task_4 train={bs:8192,steps:60} eval=320	0.49	0.25	0.86	0.76	0.77	0.63	0.43
OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.50	0.05	0.90	0.60	0.50	0.51	0.31
OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.31	0.00	0.82	0.45	0.50	0.42	0.20
OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.36	0.00	0.57	0.39	0.06	0.28	0.17
OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.50	0.05	0.90	0.60	0.50	0.51	0.31
OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.31	0.00	0.82	0.45	0.50	0.42	0.20
OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.36	0.00	0.57	0.39	0.06	0.28	0.17
OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.50	0.05	0.90	0.60	0.50	0.51	0.31
OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.31	0.00	0.82	0.45	0.50	0.42	0.20
OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.36	0.00	0.57	0.39	0.06	0.28	0.17
OPD teacher=RL task_4 train={bs:8192,steps:50} eval=312	0.01	0.01	0.03	0.02	0.03	0.02	0.01

Combined Table

Table 4: All tasks absolute metrics

Training Task	Model	Task 0	Task 1	Task 2	Task 3	Task 4	TRN AVG	HLD AVG
task_0	Base (old)	0.60	0.28	0.78	0.82	0.29	0.55	0.20
task_0	Base	0.66 ± 0.02	0.20 ± 0.02	0.89 ± 0.01	0.65 ± 0.01	0.32 ± 0.02	0.54 ± 0.01	0.40 ± 0.00
task_0	GRPO task_0 train={bs:8192,steps:10} eval=80	0.65	0.17	0.75	0.64	0.24	0.49	0.40
task_0	GRPO task_0 train={bs:8192,steps:10} eval=80	0.79	0.23	0.82	0.57	0.20	0.52	0.38
task_0	GRPO task_0 train={bs:8192,steps:10} eval=80	0.65	0.17	0.75	0.64	0.24	0.49	0.40
task_0	GRPO task_0 train={bs:8192,steps:10} eval=80	0.79	0.23	0.82	0.57	0.20	0.52	0.38
task_0	GRPO task_0 train={bs:2048,steps:50} eval=800	0.85	nan	nan	nan	nan	0.85	nan
task_0	GRPO task_0 train={bs:4096,steps:50} eval=80	0.90	0.17	0.94	0.68	0.26	0.59	0.39
task_1	Base (old)	0.60	0.28	0.78	0.82	0.29	0.55	0.20
task_1	Base	0.66 ± 0.02	0.20 ± 0.02	0.89 ± 0.01	0.65 ± 0.01	0.32 ± 0.02	0.54 ± 0.01	0.40 ± 0.00
task_1	SFT teacher eval (task 1)	0.00	0.95	0.00	0.00	0.00	0.19	0.00
task_1	GRPO task_1 train={bs:8192,steps:20} eval=320	0.62	0.27	0.94	0.79	0.30	0.59	0.35
task_1	GRPO task_1 train={bs:8192,steps:20} eval=320	0.68	0.38	0.91	0.64	0.34	0.59	0.42
task_1	GRPO task_1 train={bs:2048,steps:50} eval=800	nan	0.37	nan	nan	nan	0.37	nan
task_1	GRPO task_1 train={bs:8192,steps:20} eval=320	0.62	0.27	0.94	0.79	0.30	0.59	0.35
task_1	GRPO task_1 train={bs:8192,steps:20} eval=320	0.68	0.38	0.91	0.64	0.34	0.59	0.42
task_1	GRPO task_1 train={bs:8192,steps:40} eval=320	0.65	0.61	0.90	0.64	0.38	0.64	0.44
task_1	GRPO task_1 train={bs:8192,steps:60} eval=320	0.67	0.75	0.89	0.68	0.44	0.69	0.43
task_1	GRPO task_1 train={bs:4096,steps:50} eval=80	0.91	0.76	0.81	0.70	0.35	0.71	0.41
task_1	OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.69	0.31	0.70	0.72	0.42	0.57	0.34
task_1	OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.62	0.21	0.78	0.78	0.34	0.55	0.19
task_1	OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.57	0.46	0.70	0.68	0.09	0.50	0.13
task_1	OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.69	0.31	0.70	0.72	0.42	0.57	0.34
task_1	OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.62	0.21	0.78	0.78	0.34	0.55	0.19
task_1	OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.57	0.46	0.70	0.68	0.09	0.50	0.13
task_1	OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.69	0.31	0.70	0.72	0.42	0.57	0.34
task_1	OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.62	0.21	0.78	0.78	0.34	0.55	0.19
task_1	OPD teacher=SFT task_1 train={bs:8192,steps:50} eval=80	0.57	0.46	0.70	0.68	0.09	0.50	0.13
task_1	OPD teacher=RL task_1 train={bs:8192,steps:50} eval=80	0.70	0.61	0.70	0.81	0.31	0.63	0.34
task_4	Base (old)	0.60	0.28	0.78	0.82	0.29	0.55	0.20
task_4	Base	0.66 ± 0.02	0.20 ± 0.02	0.89 ± 0.01	0.65 ± 0.01	0.32 ± 0.02	0.54 ± 0.01	0.40 ± 0.00
task_4	SFT teacher eval (task 4)	0.01	0.00	0.09	0.15	0.87	0.22	0.02
task_4	GRPO task_4 train={bs:8192,steps:20} eval=320	0.34	0.08	0.88	0.54	0.33	0.43	0.31
task_4	GRPO task_4 train={bs:8192,steps:20} eval=320	0.53	0.28	0.87	0.72	0.61	0.60	0.39
task_4	GRPO task_4 train={bs:8192,steps:40} eval=320	0.40	0.11	0.84	0.60	0.57	0.50	0.34
task_4	GRPO task_4 train={bs:8192,steps:40} eval=320	0.53	0.29	0.88	0.72	0.70	0.62	0.41
task_4	GRPO task_4 train={bs:8192,steps:20} eval=320	0.34	0.08	0.88	0.54	0.33	0.43	0.31
task_4	GRPO task_4 train={bs:8192,steps:20} eval=320	0.53	0.28	0.87	0.72	0.61	0.60	0.39
task_4	GRPO task_4 train={bs:8192,steps:40} eval=320	0.40	0.11	0.84	0.60	0.57	0.50	0.34
task_4	GRPO task_4 train={bs:8192,steps:40} eval=320	0.53	0.29	0.88	0.72	0.70	0.62	0.41
task_4	GRPO task_4 train={bs:8192,steps:60} eval=320	0.49	0.25	0.86	0.76	0.77	0.63	0.43
task_4	OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.50	0.05	0.90	0.60	0.50	0.51	0.31

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Table 4: All tasks absolute metrics

Training Task	Model	Task 0	Task 1	Task 2	Task 3	Task 4	TRN AVG	HLD AVG
task_4	OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.31	0.00	0.82	0.45	0.50	0.42	0.20
task_4	OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.36	0.00	0.57	0.39	0.06	0.28	0.17
task_4	OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.50	0.05	0.90	0.60	0.50	0.51	0.31
task_4	OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.31	0.00	0.82	0.45	0.50	0.42	0.20
task_4	OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.36	0.00	0.57	0.39	0.06	0.28	0.17
task_4	OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.50	0.05	0.90	0.60	0.50	0.51	0.31
task_4	OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.31	0.00	0.82	0.45	0.50	0.42	0.20
task_4	OPD teacher=SFT task_4 train={bs:8192,steps:50} eval=80	0.36	0.00	0.57	0.39	0.06	0.28	0.17
task_4	OPD teacher=RL task_4 train={bs:8192,steps:50} eval=312	0.01	0.01	0.03	0.02	0.03	0.02	0.01